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Andreas Lydakis

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Software engineer with over 10 years of experience in the AV industry, HRI, robotics. Experienced in integrating complete systems, V&V and proven track record of working with interdisciplinary teams as a team leader and product owner.

SKILLS

Simulation Frameworks	ROS/ROS2, RtiDDS, Applied Intuition Tools, experience with Carla and NVIDIA Isaac
Data Science & ML	Pandas/Scikit, experience with Databricks, familiarity with Tensorflow/Pytorch, certification on Watson IBM Watson
Programming Languages	C++, Python, R, bash scripting, exposure to Rust/Kotlin
Tools & CI	Git, CMake, Bazel
Game Engines	Unity, Unreal Engine through personal projects

TECHNICAL EXPERIENCE

Senior SWE/ S/W Architect **Sept 2021 — Present**
Scania Group *Sodertalje, Sweden*

- Lead the implementation and integration of different simulation tools, their interactions with the onboard stack, data storage pipelines, and the publishing and tracking of KPIs for different application domains.
- Participated in interteam design of V&V pipelines, test suite orchestration and the necessary infrastructure to track requirements when running tests.
- Personally wrote and maintained local development tools to assist every day development
- Participated in code reviews, onboarding new team members

Product Owner - Simulation **Sept 2022 — Present**
Scania Group *Sodertalje, Sweden*

- Lead a team of 9 responsible for onboard and offboard simulation.
- Communicated with development teams and stakeholders, gathering requirements and coordinating quarterly development cycles for three distinct products.
- Coordinated supplier and tool evaluation in cooperation with development team stakeholders.

Robotics Engineer **October 2018 - August 2021**
Dyson Technology *Malmesbury, UK*

- Lead the design and implementation a multi-modal distributed HRI behavioural model, based on behaviour trees, demonstrated on hardware prototypes.
- Worked on human gesture recognition, and DL-based human-robot demonstration applications.
- Designed simulated manipulation and computer vision applications, and testing frameworks.
- Implemented robotics navigation components, and related evaluation and testing frameworks.
- Worked on migrating the team's robotics stack from ROS to ROS2.

Robotics Engineer **May 2018 - August 2018**
Innotec *Cambridge, UK*

- Implemented ROS nodes for CAN and RS485 based communication and visual odometry systems for wind turbine maintenance robot projects.
- Implemented a proof-of-concept motion capture based system for activity tracking and evaluation.

Research Assistant **Sept 2015 - May 2018**
University Of New Hampshire *Durham, NH, USA*

- Developed RL based wearable robotics applications for personalized upper and lower limb rehabilitation
- Published work in Human-Robot Interaction, Robust Optimization Techniques.

Research Engineer **June 2013 - August 2015**
NCSR Demokritos *Athens, Greece*

- Developed ROS applications in C++ & Python for autonomous and remote controlled robot navigation, robot vision, interfacing with Android devices, and communication with Raspberry Pi and Arduino controlled devices.
- Participated in outreach initiatives for university students and grade-school children.

EDUCATION

MSc, Computer Science , <i>University of New Hampshire</i>	May 2018
Diploma, Computer Engineering & Informatics , <i>University of Patras</i>	May 2013

CERTIFICATIONS/COURSES

IBM - AI Engineering
IBM - Generative AI Engineering
Duke University - MLOps
Deep Learning.AI - Deep Learning Specialization
Deep Learning.AI - TensorFlow Advanced Techniques
LinkedIn - Scrum: Advanced

PUBLICATIONS

IJCAI-ECAI-18: Artificial Intelligence for Wildlife Conservation (AIWC) workshop. Computing Robust Strategies for Managing Invasive Plant Species. Lydakis A. , Allen J., Petrik M., Szewczyk T.	July 2018
15TH International Conference on Rehabilitation Robotics. A Learning-based Agent for Home Neurorehabilitation. Lydakis A. , Yuanliang M., Munroe C.	July 2017
International Conference on Pervasive Technologies Related to Assistive Environments. Irregular Gait Detection using Wearable Sensors. Lydakis A. , Kao P.C., Begum M.	June 2017
EAI Endorsed Transactions on Pervasive Health and Technology , Special Issue on Mobile and Wireless Technologies for Healthcare. Interacting with and via mobile devices and mobile robots in an assisted living setting. Dagigioglou M., Lydakis A. , Kirstein F., Dogruoz A.S, Konstantopoulos S.	February 2015
9th Annual Conference on Human-Robot Interaction (HRI 2014) Embodied Visual Programming for Robot Control Konstantopoulos S., Lydakis A. , Gkikakis A.E.	March 2014